

## 1. Patient database (EHR) And data ingestion

Structured data (patient records)+  
unstructured data (clinic notes: pdf)

SOAP Notes: ICD-10 Codes (E11.9 =  
Type 2 diabetes)

Relational DB schema design  
SQLite

Parse PDFs and then ingest them into our  
vector DB.

STORE PATIENT RECORDS  
DB Schema: 2 tables: patients,  
medical\_records

Pypdfloader  
Pdfreader

## Milestone 2: Doctor Schedule API

TABLES for our database

doctors (name, speciality, location/city)

appointments (patient, doctor, slot,  
reason, status, booked\_by, run\_id)

slots (doctor, start, end, status)

Seed doctors in to our database: 11

doctors, 9 specialities

30 min slot,

Prefer doctors in patient's city. Ex. If a  
person Ramesh stays in Chennai he  
should book an appointment in his  
location Chennai

If someone books an appointment: slot:  
booked for doctor

If someone cancels an appointment: slot  
should be available for booking by other  
patients.

**Milestone 3: LLM Layer and structured  
prompts:**

Prompt template

ChatPromptTemplate: system and human  
message, variable,

To make sure our output is more structured: PydanticModel: to make chaining using lang chain more reliable.

Prompt chaining: profile and records and memories —> summary —> render info

Do we want to show the appointment summary/history?

Prompt: specific to each tasks

**Milestone 4: Vector db+memory —> FAISS**

1. Kidney problems/CKD Stage 3:  
semantic search/similarity search

Memory strategy:

Short -term: chat turn

Long-term: (facts learned across our sessions)

Records: EHR, source of truth

Metadata filters

Privacy rule: never mix another patient's memory into a prompt.

## Milestone 5: RAG Medical search (medlinePlus +WHO)

Fetch —> clean —> chunk —> embed —  
> retrieve —> generate with citations

Trusted sources: MedlinePlus websearch  
WHO Fact sheet

Responses should be grounded and  
provided with citations (source of truth)

## Chunking strategy:

RecursiveCharacterTextSplitter

RETRIEVE TOP K CHUNKS FOR  
questions.

## Milestone 6: Planner and goal decomposition

Plan everything first and then will execute  
Guardrails:

## Milestone 7: CrewAI/AutoGen/ StrandsAgent SDK/Langgraph: Agentic AI Framework: Langgraph

## Milestone 8: Evaluation with QAEvalChain:

LLM As a judge Gold question bank and answers: will compare predicted answers with gold answers and give rating

## Milestone 9: streamline dashboard, memory, logs